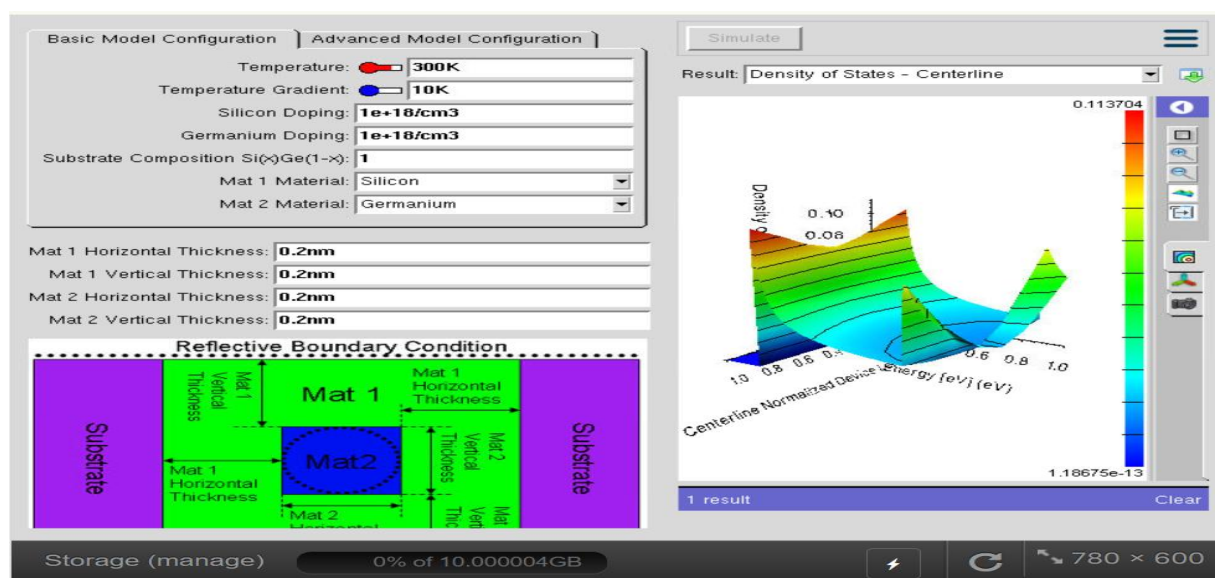


Exp. 2. Simulation of Density of States of Quantum Well Structure with Various Nanometers Thickness.

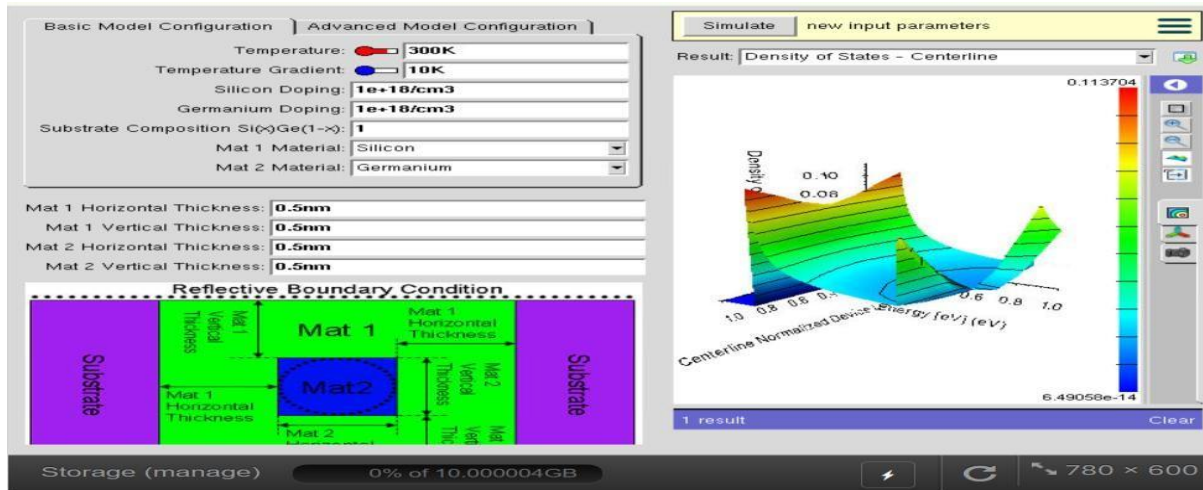
Aim: To simulate density of states of quantum well structure with various nanometer thickness.

Output:



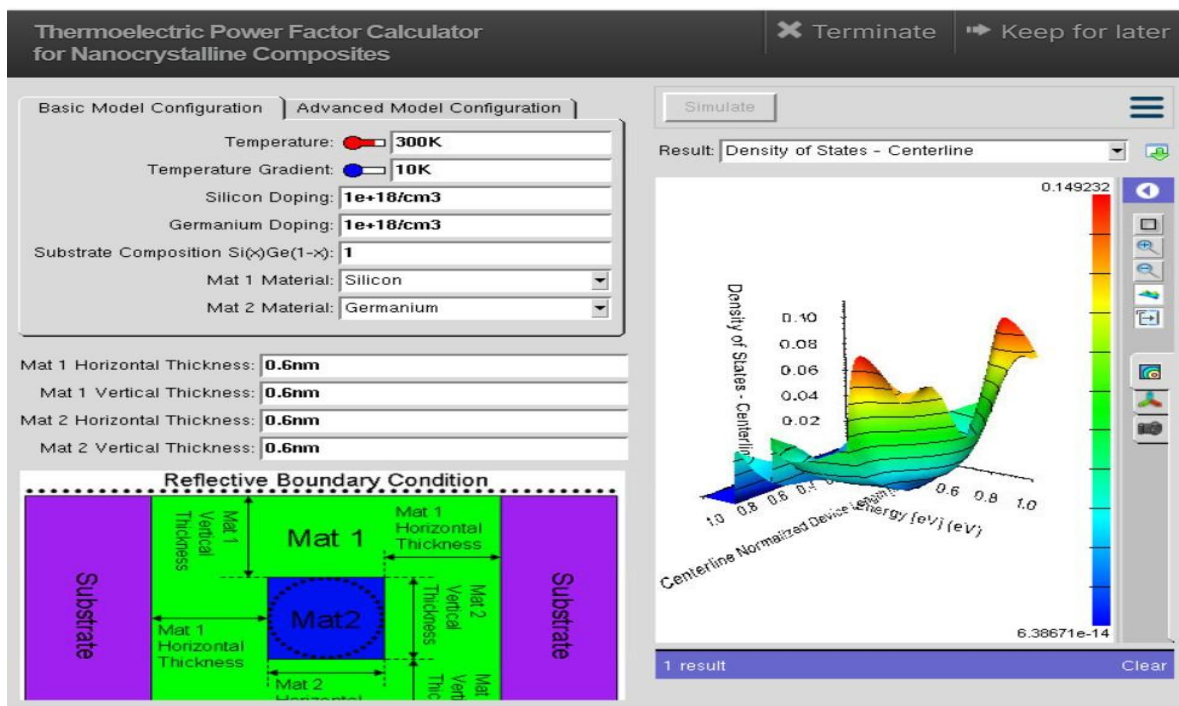
The graph shows the Density of States (DOS) at the centerline of a Silicon–Germanium nanocomposite with a thickness of 0.2 nm at 300 K. The DOS exhibits relatively low values with small peaks due to strong quantum confinement at the nanoscale. The limited number of available energy states indicates restricted electron movement, which influences the electrical and thermoelectric properties of the nanocomposite.

Case Study 1: Simulate the density of states of quantum well of thickness 0.5 nm



Output: The graph shows the Density of States (DOS) at the centerline of a Silicon–Germanium nanocomposite with a thickness of 0.5 nm at 300 K. The DOS increases with energy and exhibits distinct peaks due to quantum confinement, indicating the availability of more electron energy states. Compared to thinner structures, the 0.5 nm thickness provides a higher density of states, resulting in improved carrier transport and enhanced thermoelectric performance.

Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm



Output: The graph shows the Density of States (DOS) at the centerline of a Silicon–Germanium nanocomposite with a thickness of 0.6 nm at 300 K. The DOS exhibits higher and sharper peaks than the 0.5 nm structure, indicating an increase in the number of available electron energy states. The increased thickness reduces quantum confinement, improving carrier transport and

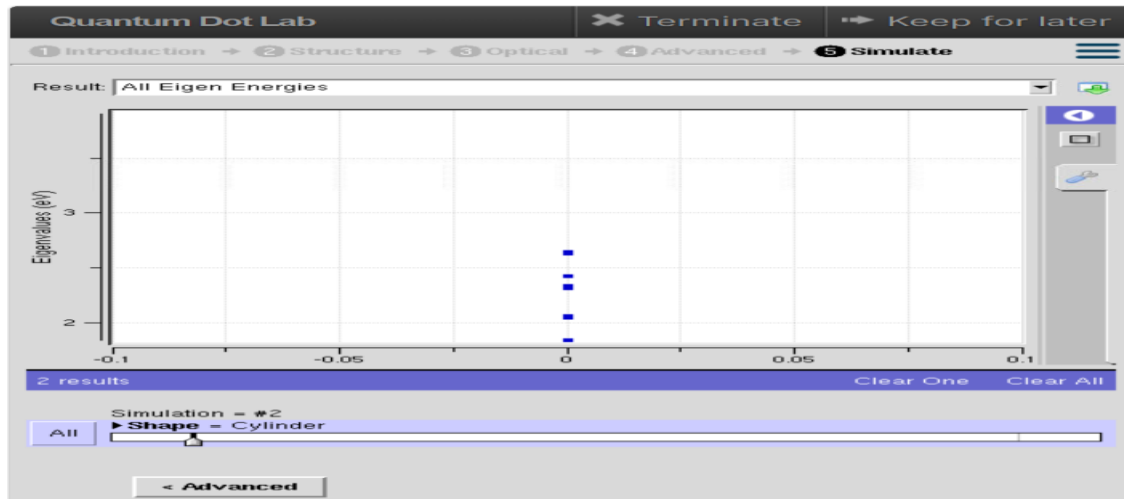
enhancing the thermoelectric performance of the nanocomposite.

Result: Simulation of density of states of quantum well structure with various nanometer thickness is accomplished.

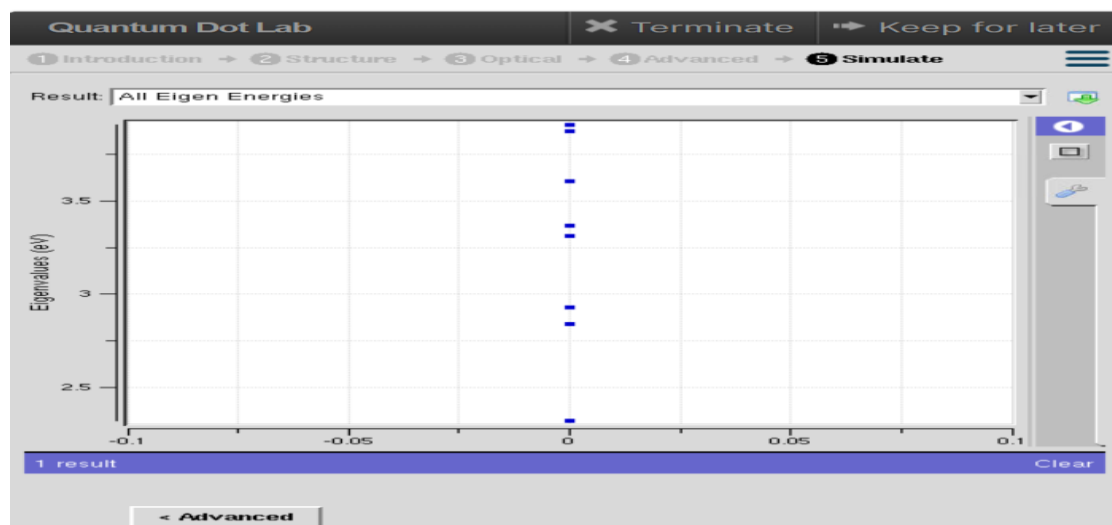
Exp. 3. Simulation of Energy States of Quantum Dot using Nano Hub

Aim: To obtain the energy states of Quantum dots.

Output:

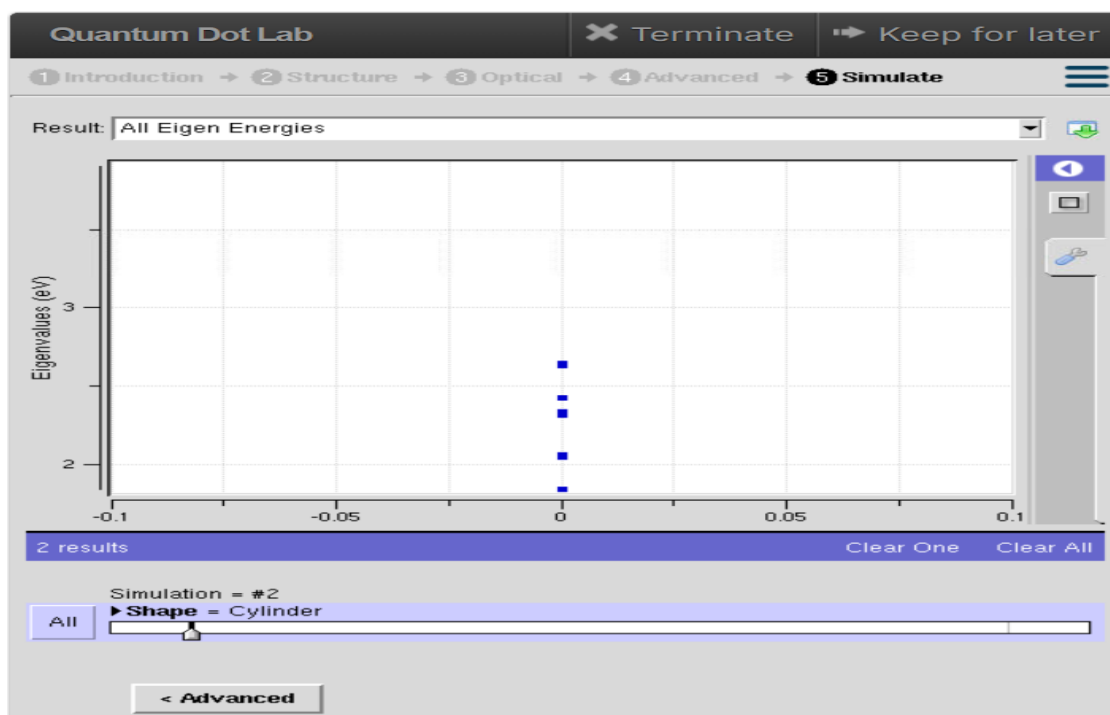


Case Study 1: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8 nm.



Output: The graph shows the eigen energy levels of a quantum dot. The discrete blue points indicate that electrons can occupy only specific energy levels due to quantum confinement. The energy levels are well separated, confirming the quantized nature of the quantum dot. This discrete energy spectrum is important for improving the optical and electronic properties of quantum dot devices.

Case Study 2: Simulate the energy states of two layer cuboid quantum dots



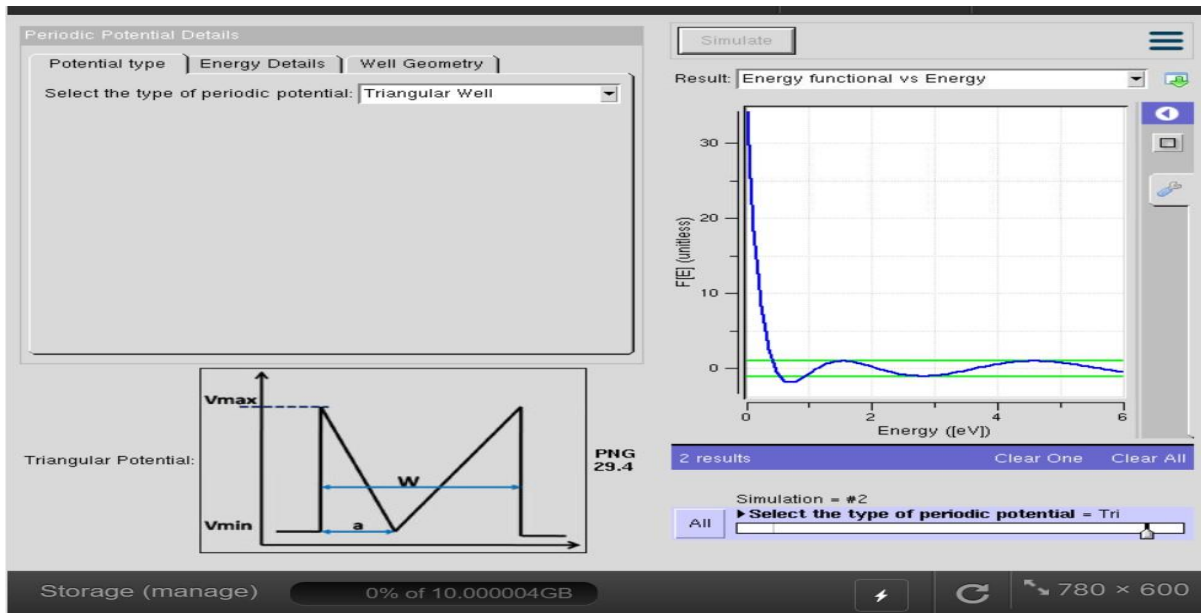
Output:

The graph shows the eigen energy levels of a quantum dot. The discrete blue points indicate that electrons can occupy only specific energy levels due to quantum confinement. The energy levels are well separated, confirming the quantized nature of the quantum dot. This discrete energy spectrum is important for improving the optical and electronic properties of quantum dot devices.

Result: The Simulation of energy states of quantum dots is accomplished.

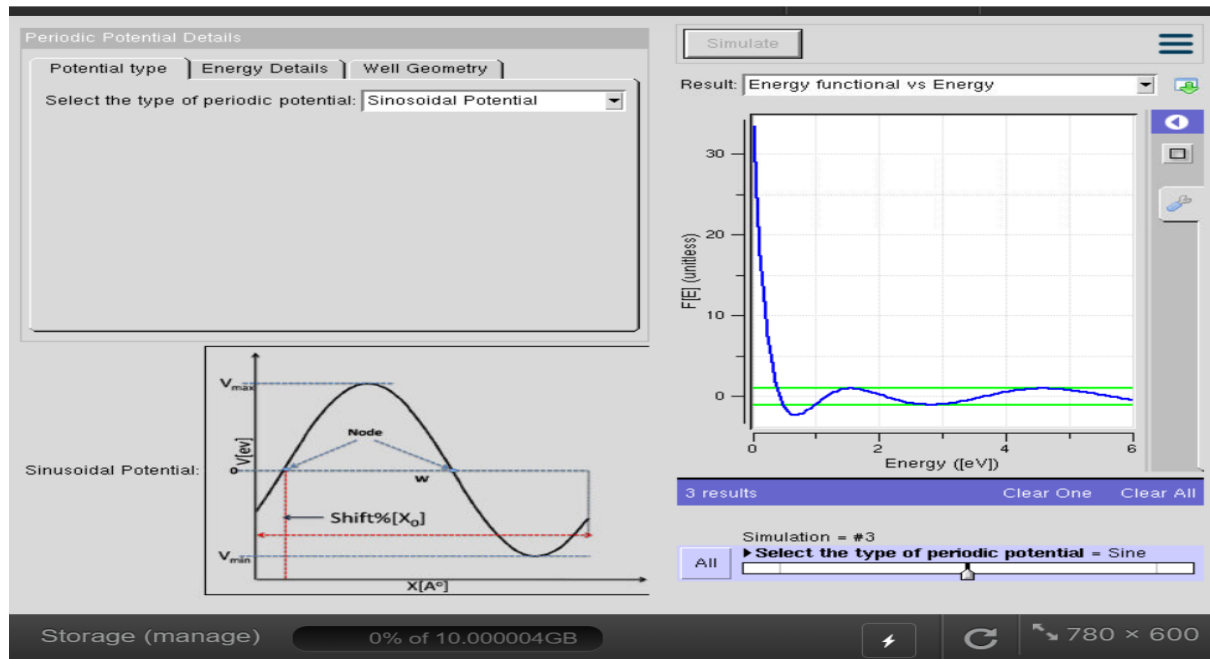
Exp. 4. Kronig Penney Model

Case Study 1: Simulate the output for Triangular well



Output: The graph shows the Energy Functional vs Energy for a triangular well periodic potential. The energy functional decreases sharply at low energy and then exhibits oscillations as the energy increases. These oscillations indicate the formation of allowed and forbidden energy bands due to the periodic triangular potential, demonstrating the quantum behavior of electrons in the structure.

Case Study 2: Simulate the output for Sinusoidal Potential



Output:

The graph shows the Energy Functional vs Energy for a sinusoidal periodic potential. The energy functional decreases rapidly at lower energies and then varies smoothly with oscillations as the energy increases. The sinusoidal potential produces smoother energy band formation, indicating periodic electron motion and efficient wave propagation through the crystal lattice.

Result: The energy functional vs Energy graph using the Kronig Penney model is simulated.